

Reproducibility of Earth System Software for Weather and Climate Analytics

Challenges, Practices, and Pathways Toward Sustainable Science

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TEXAS TECH
UNIVERSITY.



National Oceanic and
Atmospheric Administration
U.S. Department of Commerce

Background

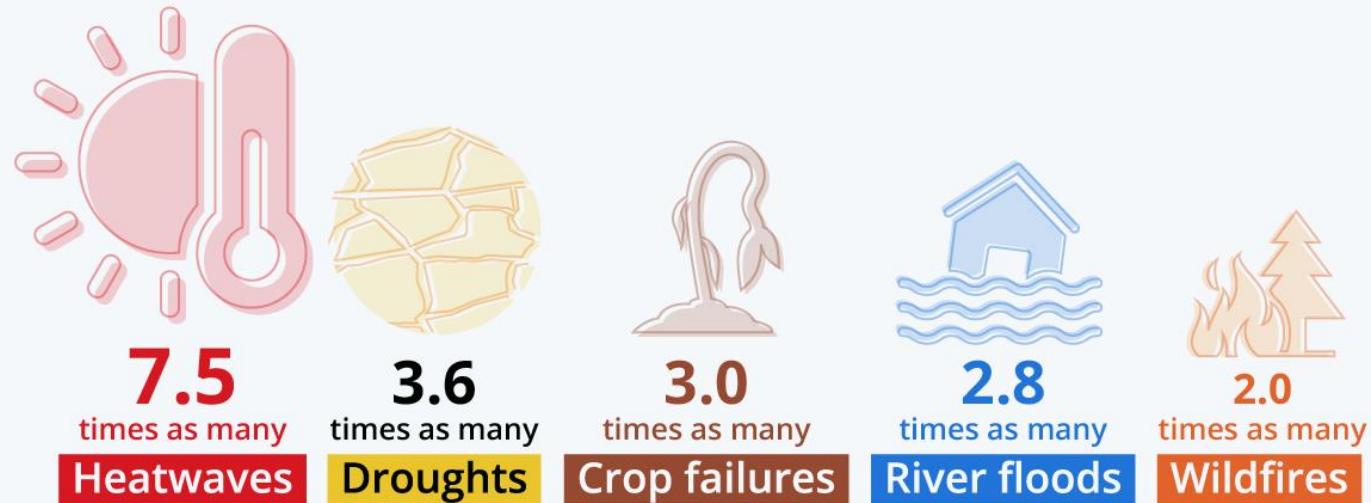
- **Reproducibility is essential for scientific integrity and progress.**
- **Increasing complexity in Earth system software and datasets.**
- **Growing concern over reproducibility gaps in weather and climate analytics.**



Motivation: Spreading of Extreme Weather Events

Number of Climate Disasters to Triple for New Generation

Frequency of climate disasters experienced in a lifetime for a person born in 2020 compared to one born in 1960



All climate disasters
~3 times as many

Based on NDC scenario (following Paris Agreement) of 2.7 °C/4.9 °F warming until 2100
Source: Thiery et al. Intergenerational Inequities in Exposure to Climate Change. Science (2021) via media reports



statista

World Sees Record Heat Waves

Selection of temperature records by country (or continent) recorded during the last five years, in °C



* As of May 3, 2022.

Sources: World Meteorological Organization, media reports, Statista research

“ The climate and inequality crises need to be addressed at the same time so that no one is left behind.

WMO Secretary-General, Prof. Celeste Saulo



statista

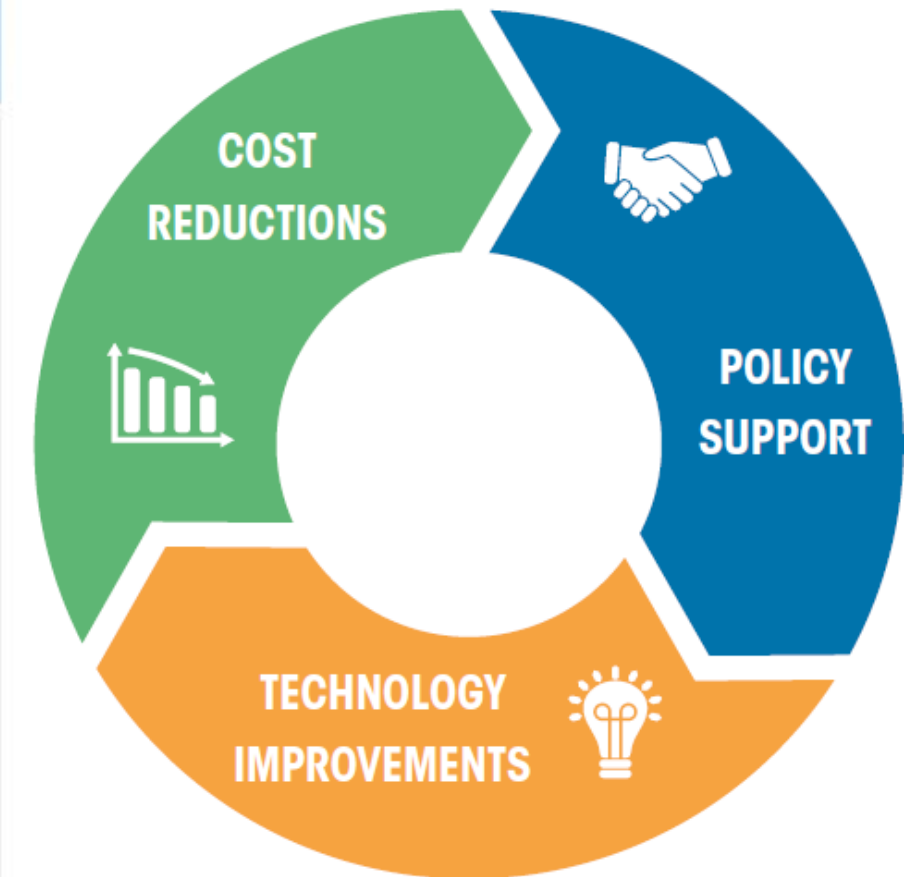
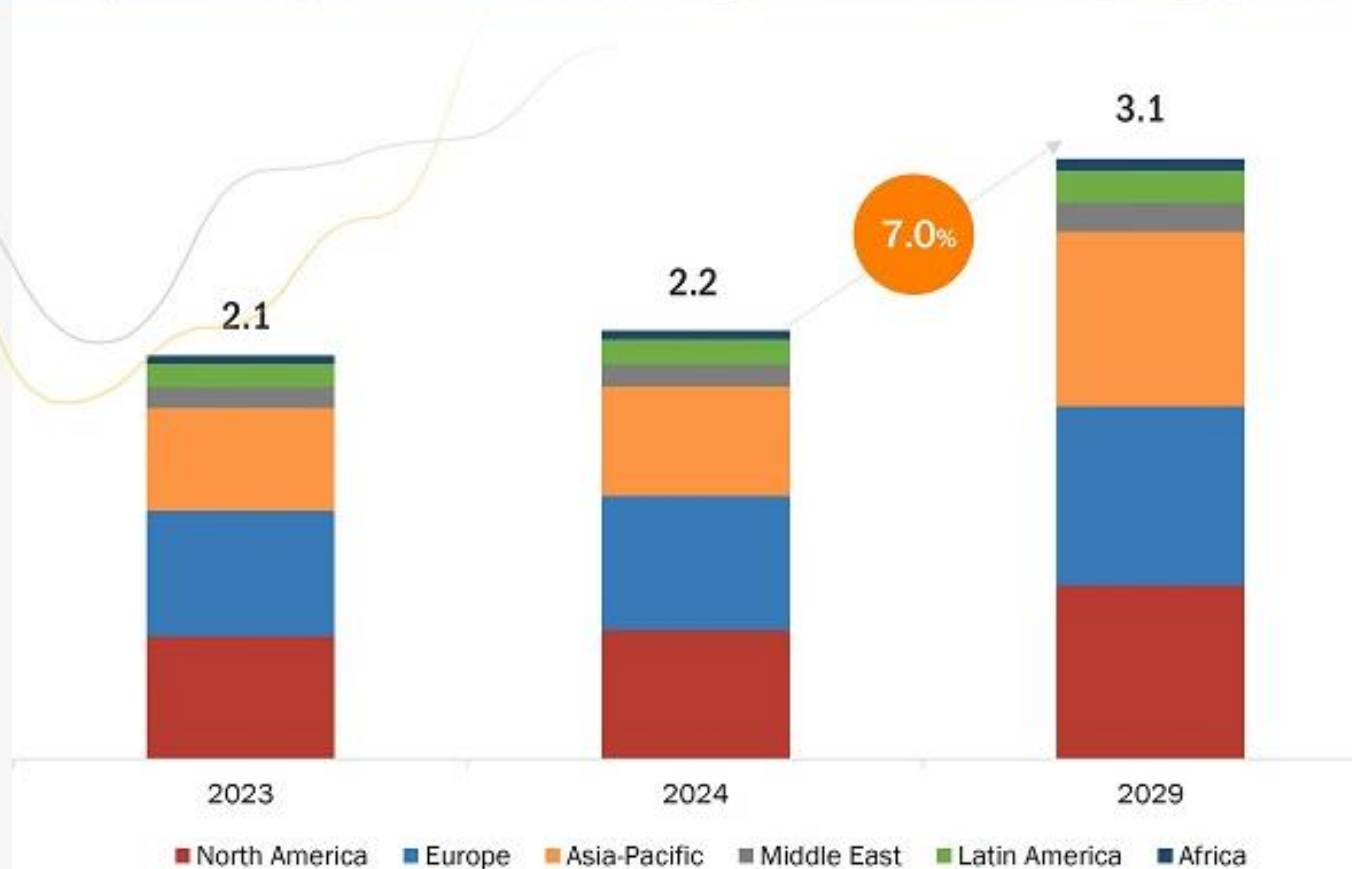
Global Investment In Weather Forecasting Systems Market

WEATHER FORECASTING SYSTEMS MARKET GLOBAL FORECAST TO 2029 (USD BN)



CAGR OF
7.0%

The global weather forecasting systems market is projected to reach USD 3.1 billion by 2029, growing at a CAGR of 7.0% from 2024 to 2029.



<https://www.marketsandmarkets.com/>

Requirements of Weather Research and Forecast



Analyze the impact of extreme weather events across various sectors.



Examine the potential of sustainable systems.



Simulation and sensitivity analysis to support the planning of sustainable systems.

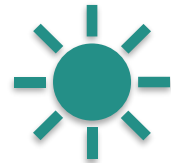
Industry Requirements



Energy
Providers and
Grid Operators



Urban Planners
and
Infrastructure
Developers



Renewable
Energy Sector



Building and
Construction
Industry

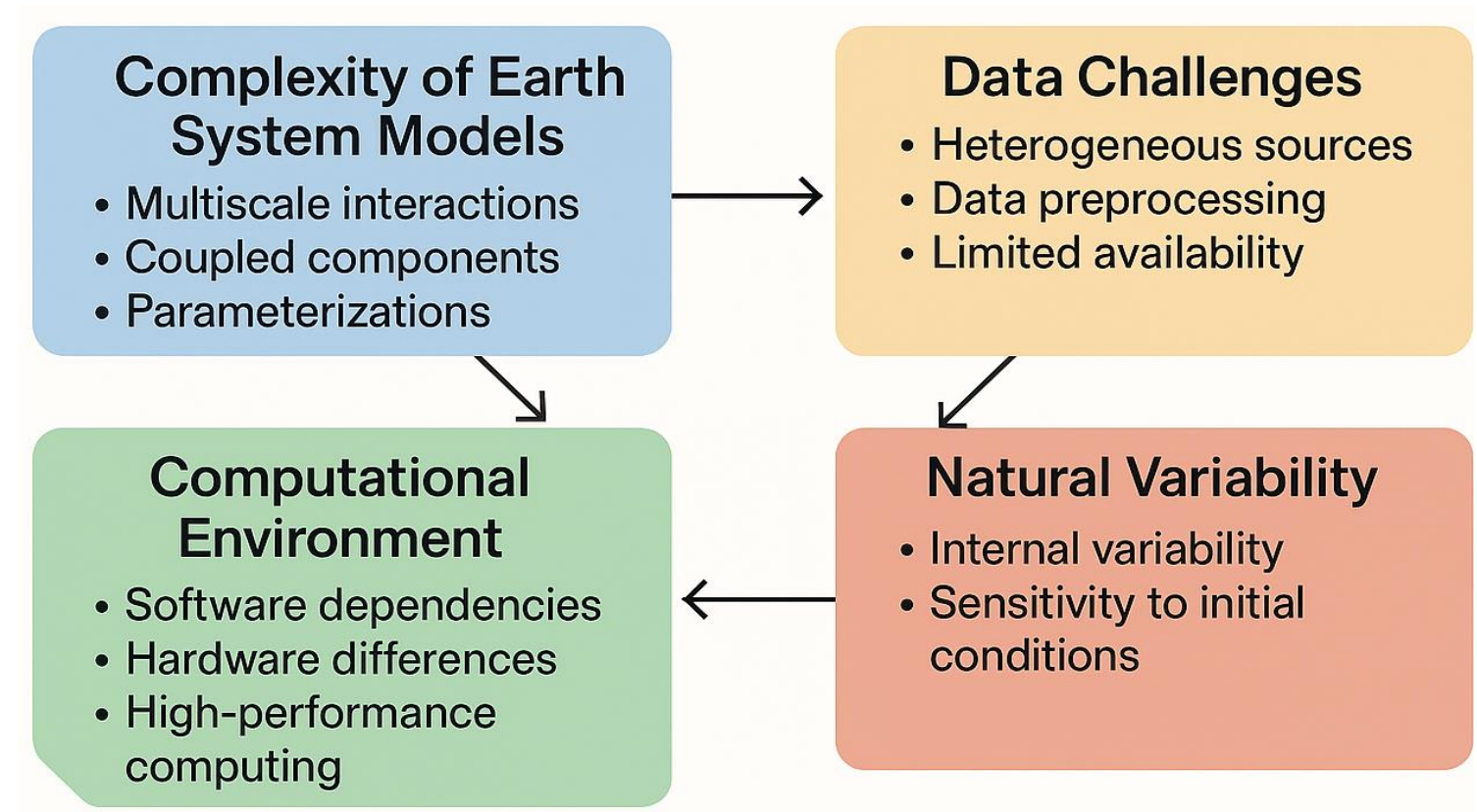
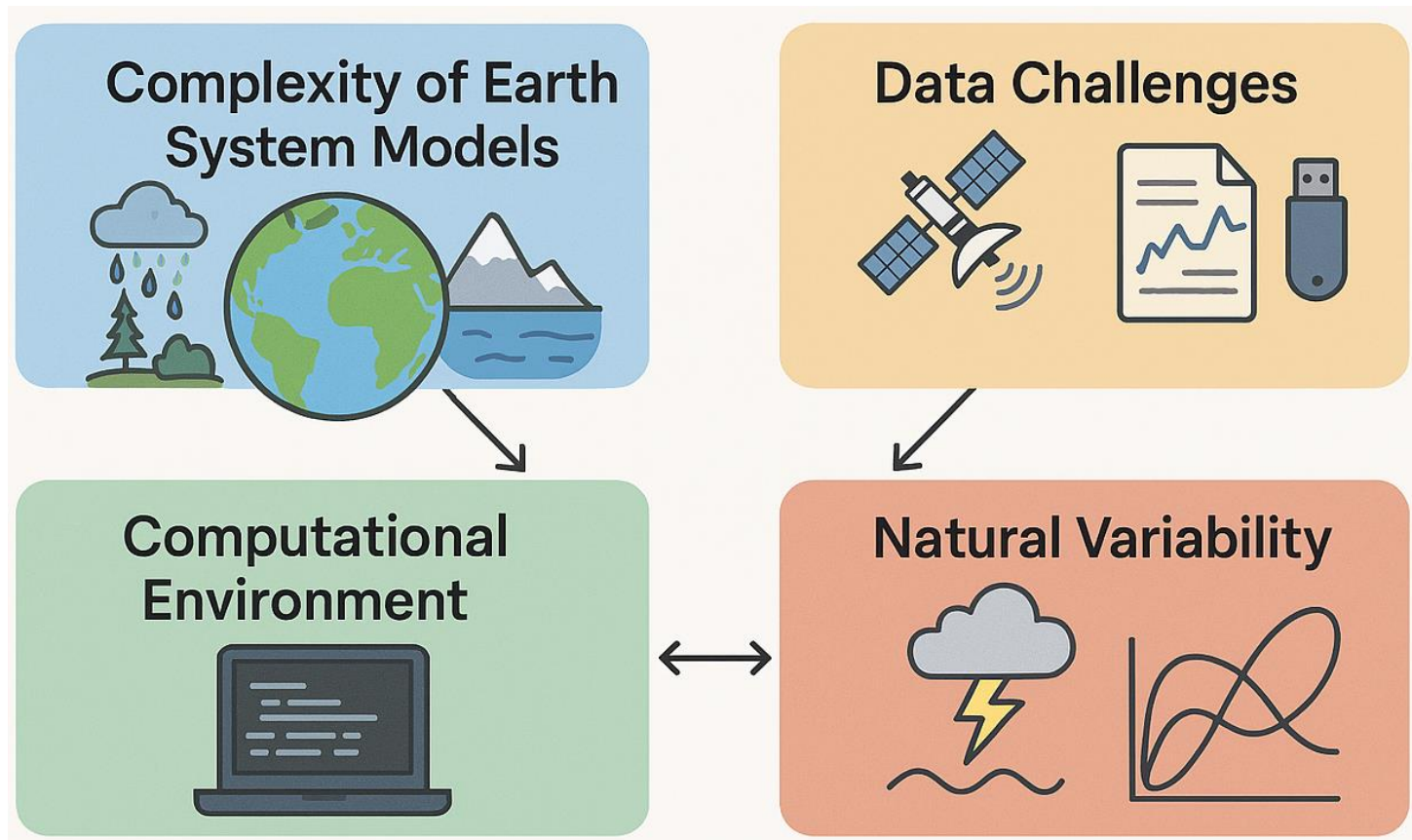


Government
and Policy
Agencies

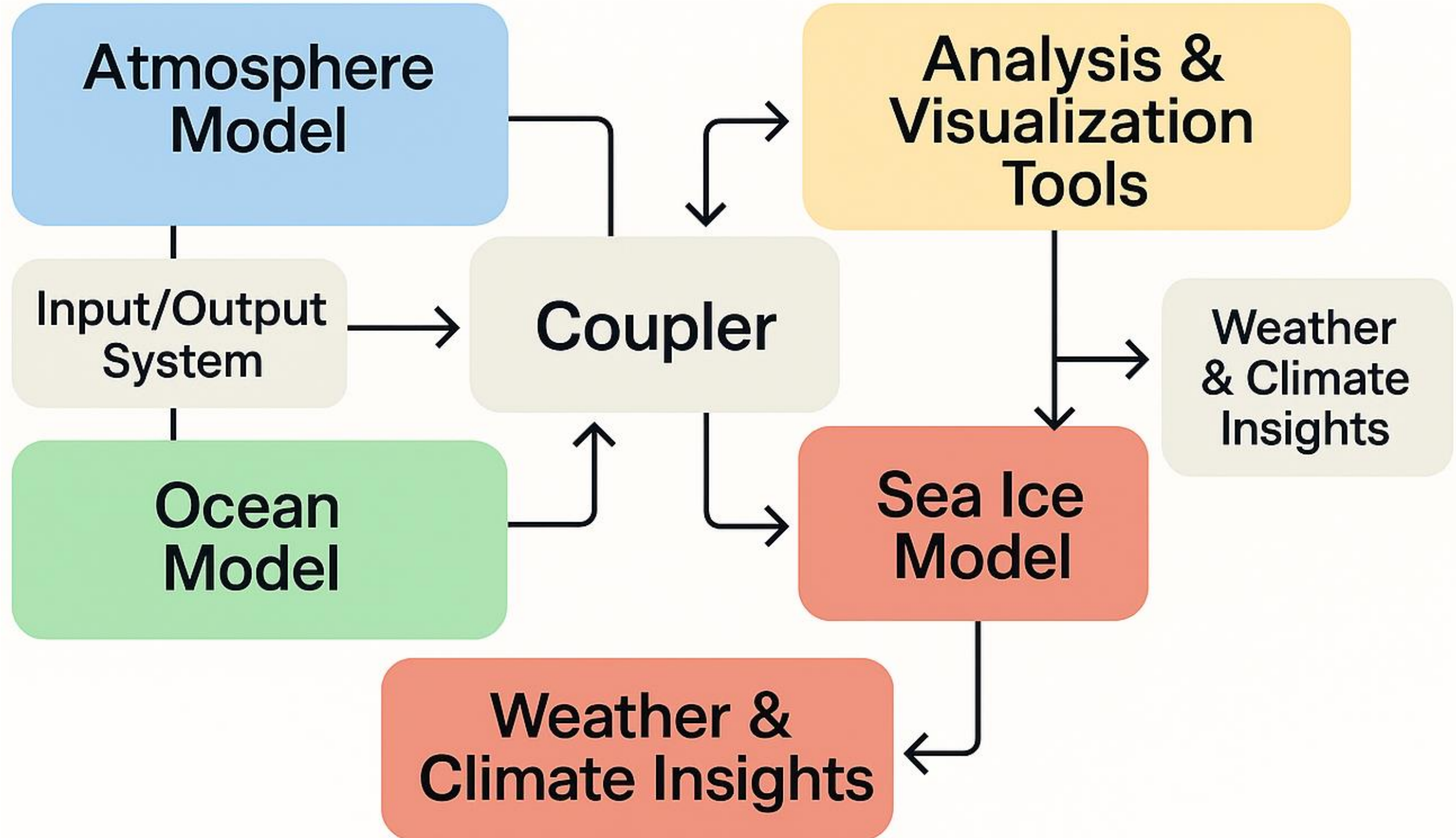


Emergency
Services and
Public Health

Why Reproducibility Is Difficult In Earth System Science



Components of ESS for Weather and Climate Analytics



Why Reproducibility Matters

**Verification &
Transparency**

**Computational
Research**

Accuracy

**Process
Integrity**

Verification & Transparency



Reproducibility Importance

Reproducibility allows independent researchers to confirm findings, enhancing credibility and reliability.

Transparency in Methodology

Open data and clear methodologies enable thorough scrutiny and reduce risks of errors or fraud.

Collaboration Through Sharing

Sharing code, data, and workflows fosters collaboration and trust among researchers.

Computational Research



Role in Modern Science

Computational research enables complex simulations and data modelling beyond manual capabilities.

Importance of Reproducibility

Reproducibility ensures consistent algorithm results across environments and datasets.

Applications in Science

Computational methods impact genomics, climate science, and artificial intelligence decision-making.

Enhancing Research Quality

Reproducible workflows help validate findings, share methods, and refine models for accuracy.

Accuracy



Role of Accuracy

Accuracy ensures scientific results are precise and reflect true observations for reliable conclusions.

Reproducibility Safeguard

Reproducibility allows independent verification of results, acting as a safeguard against inaccuracies in research.

Impact of Inaccuracy

Inaccurate results can mislead research, waste resources, and cause harm in fields like medicine and policy.

Building Knowledge Reliably

Accurate and reproducible research builds cumulative knowledge contributing reliably to broader understanding.

Process Integrity



Consistency and Reliability

Process integrity ensures scientific methods are consistent, reliable, and free from bias to maintain research quality.

Reproducibility Importance

Reproducibility requires well-documented and standardized processes to enable replication and verification of results.

Ethical Standards and Innovation

Upholding process integrity maintains ethical standards and fosters innovation by building research confidence.

Challenges in Reproducibility



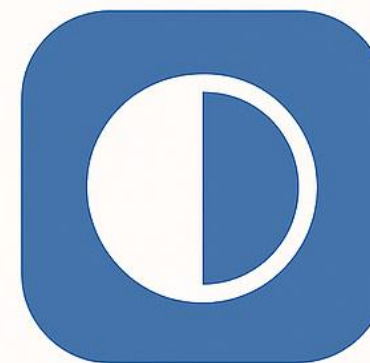
- **Variations in computational environments (hardware, software versions).**
- **Lack of comprehensive documentation and version control.**
- **Dependence on proprietary software and data.**

Current State of Reproducibility in Earth System Software

- **Fragmented Practices:** Different research groups use varying standards for documentation and version control.
- **Limited Adoption of FAIR Principles:** Many Earth system models lack full compliance with Findable, Accessible, Interoperable, and Reusable data principles.
- **Tool Diversity:** Use of multiple modeling frameworks (WRF, CESM, ESMValTool) without standardized reproducibility workflows.
- **Challenges:** Dependency mismatches, undocumented scripts, and platform-specific builds remain common barriers.



Fragmented
Standards



Limited FAIR
Compliance



Tool
Diversity



Challenges

Earth System Software for Weather and Climate Analytics



WRF



CESM



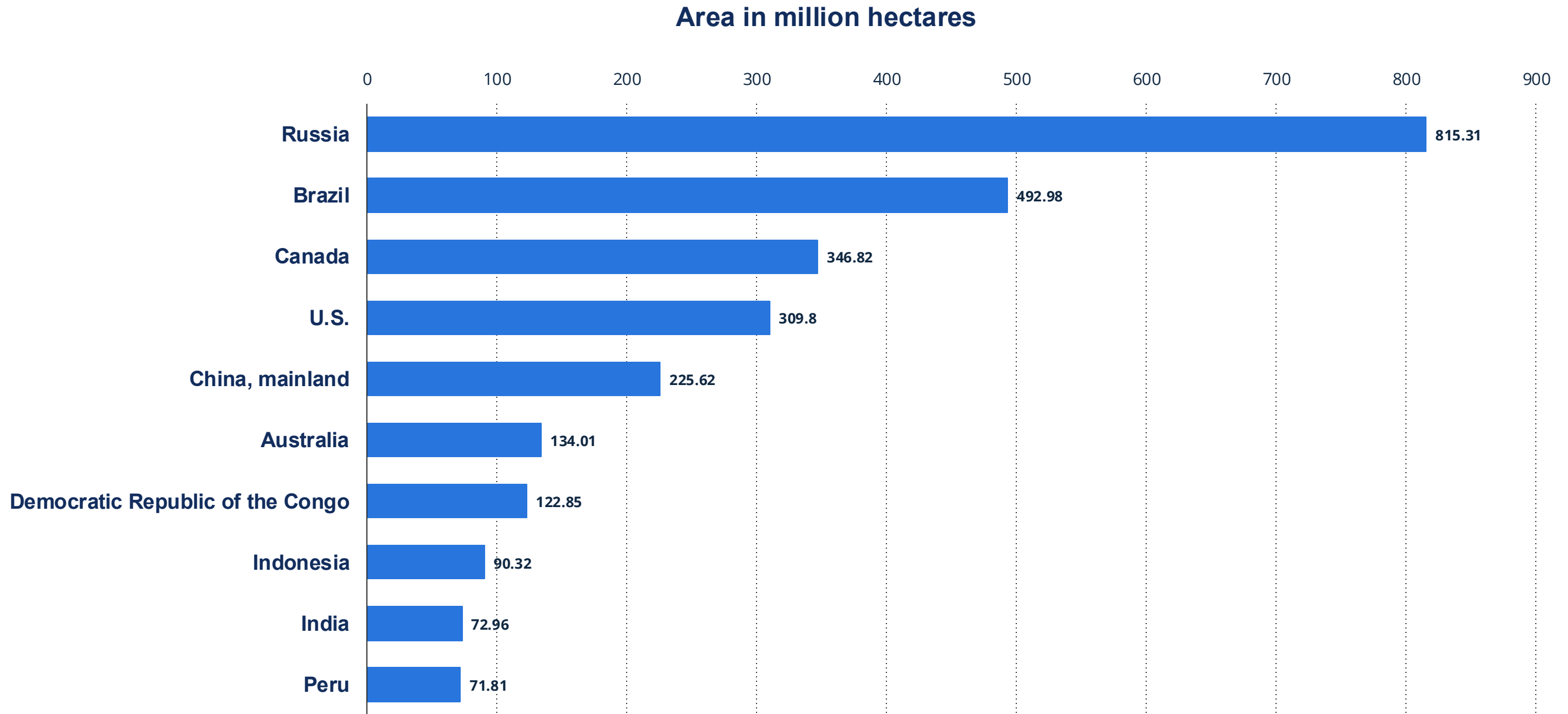
ESMValTool

WRF: Weather Research and Forecasting Model

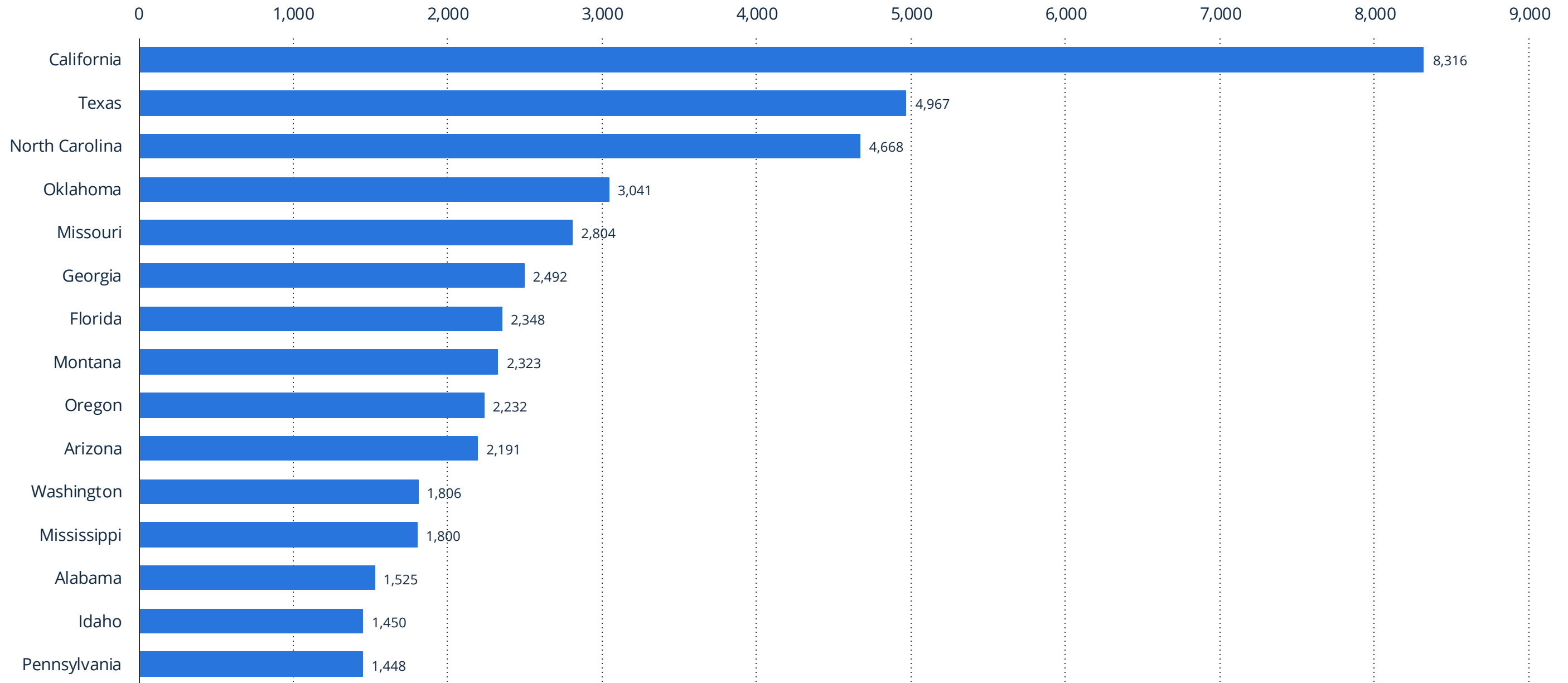
CESM: Community Earth System Model

ESMValTool: Earth System Model Evaluation Tool

Ten countries with the largest forest area in 2023 (in million hectares)



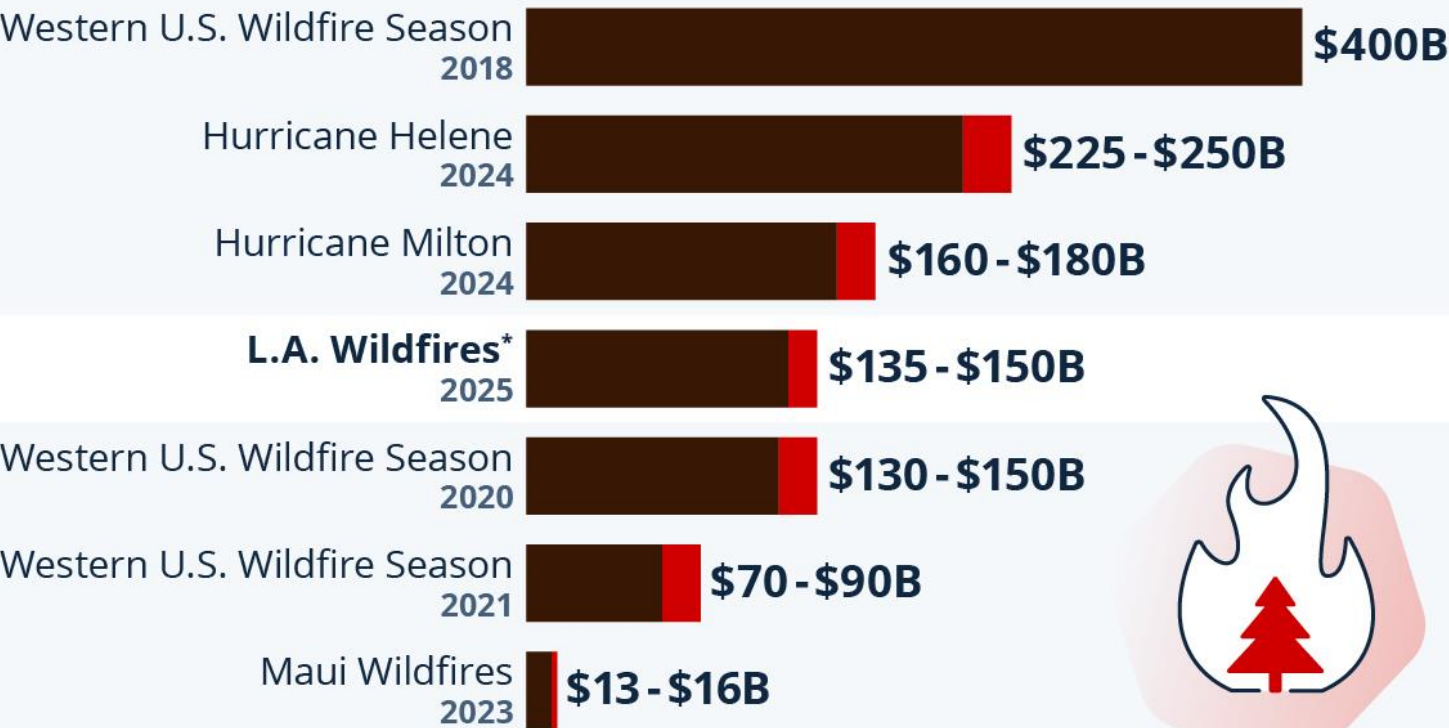
Number of wildfires in the United States in 2024, by state



Escalating Human and Economic Impact

L.A. Wildfires Among the Costliest in Recent History

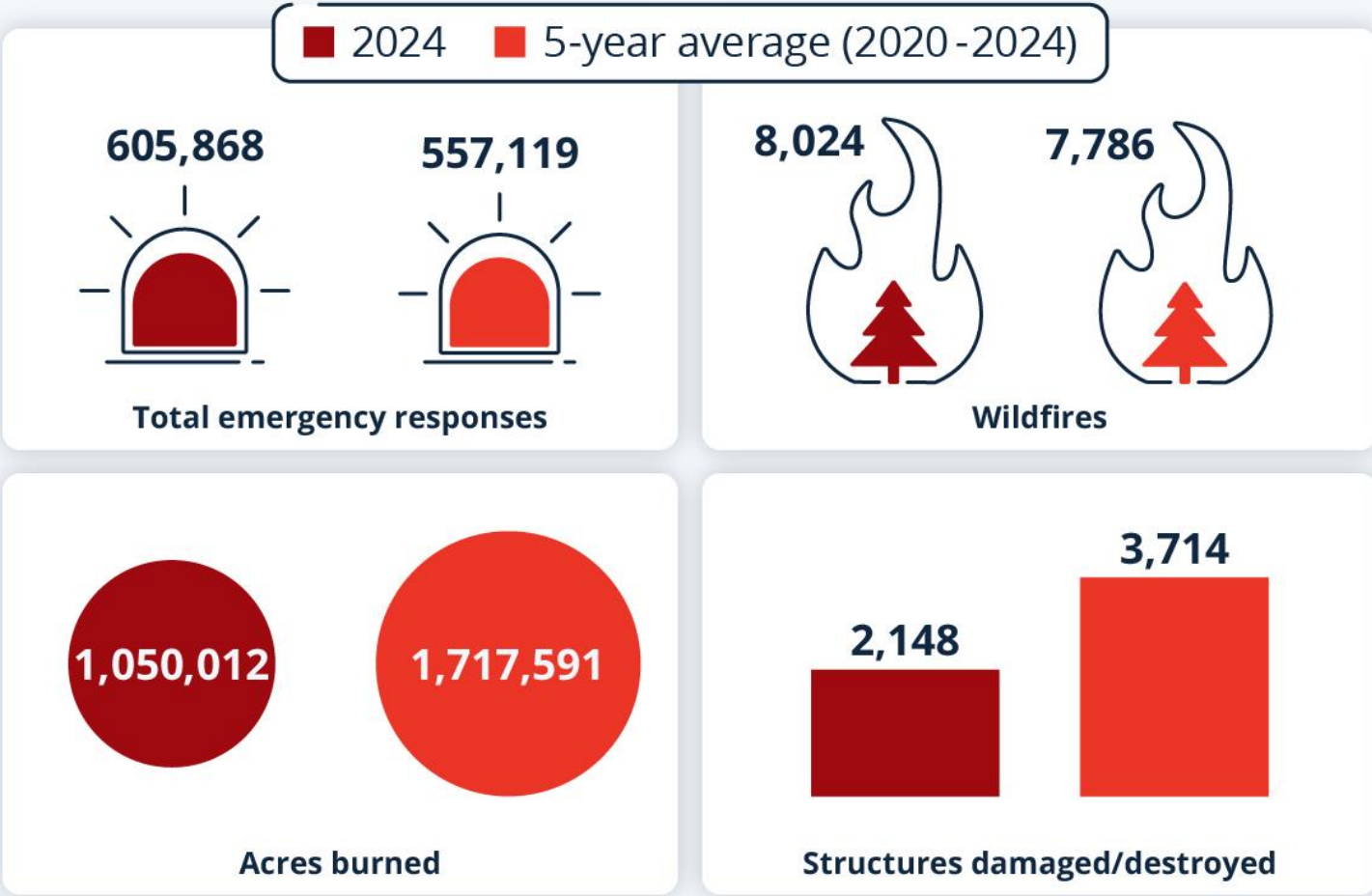
Estimated total damage and economic losses associated with recent natural disasters in the U.S.



* Latest preliminary estimate as of Jan. 9, 2025. Estimates includes direct costs (e.g. property damage) and indirect costs (e.g. wage losses and supply chain disruptions)
Source: AccuWeather

California's Wildfire Problem

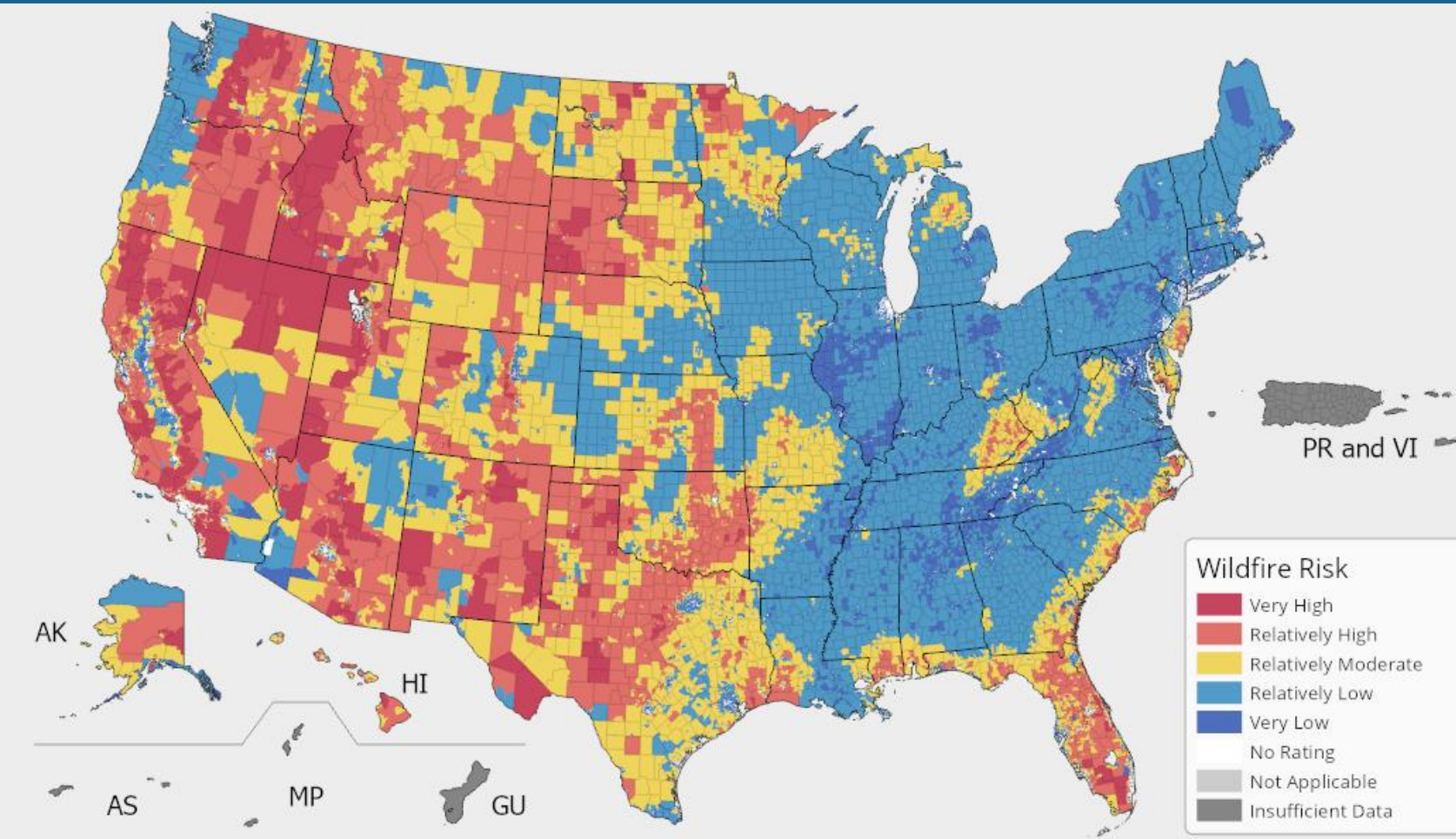
Key statistics for emergency responses connected to wildfires in California



Source: CAL FIRE

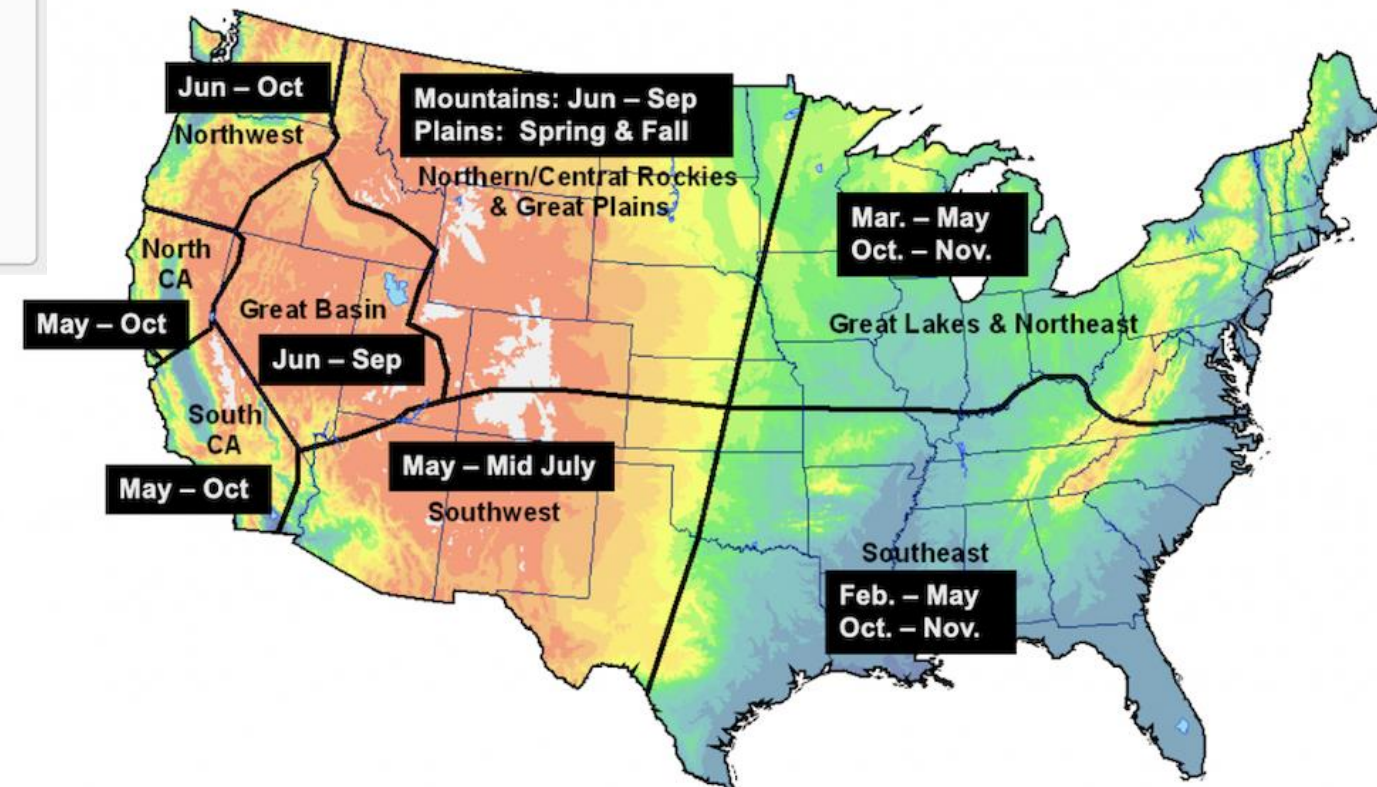
Beyond these numbers, wildfire smoke now impacts air quality for millions of people across continents, creating widespread public health concerns even for those far from actual flames.

Mapping Fire Risk with Weather and Climate Models



<https://hazards.fema.gov/nri/wildfire>

Peak Regional Fire Seasons



Alaska (not shown): May - August

<https://www.clarity.io/>

Wildfire Climate Connection: The Core Catalyst



Rising Temperatures

Global average temperature up $\sim 1.1^{\circ}\text{C}$ since pre-industrial era, with Western US temperatures rising $2.5\text{--}4.5^{\circ}\text{F}$ since the 1970s



Increased Drought

Western US experiencing its worst mega-drought in 1,200 years (2021), creating critically dry conditions



Extended Fire Seasons

US fire season now 78 days longer than in 1970, allowing more time for destructive events



Fuel Aridity

Drier vegetation with lower moisture content dramatically increases ignitability and fire intensity

Process Involved in 3km Real-Time WRF Modelling System

Data Acquisition



Preprocessing



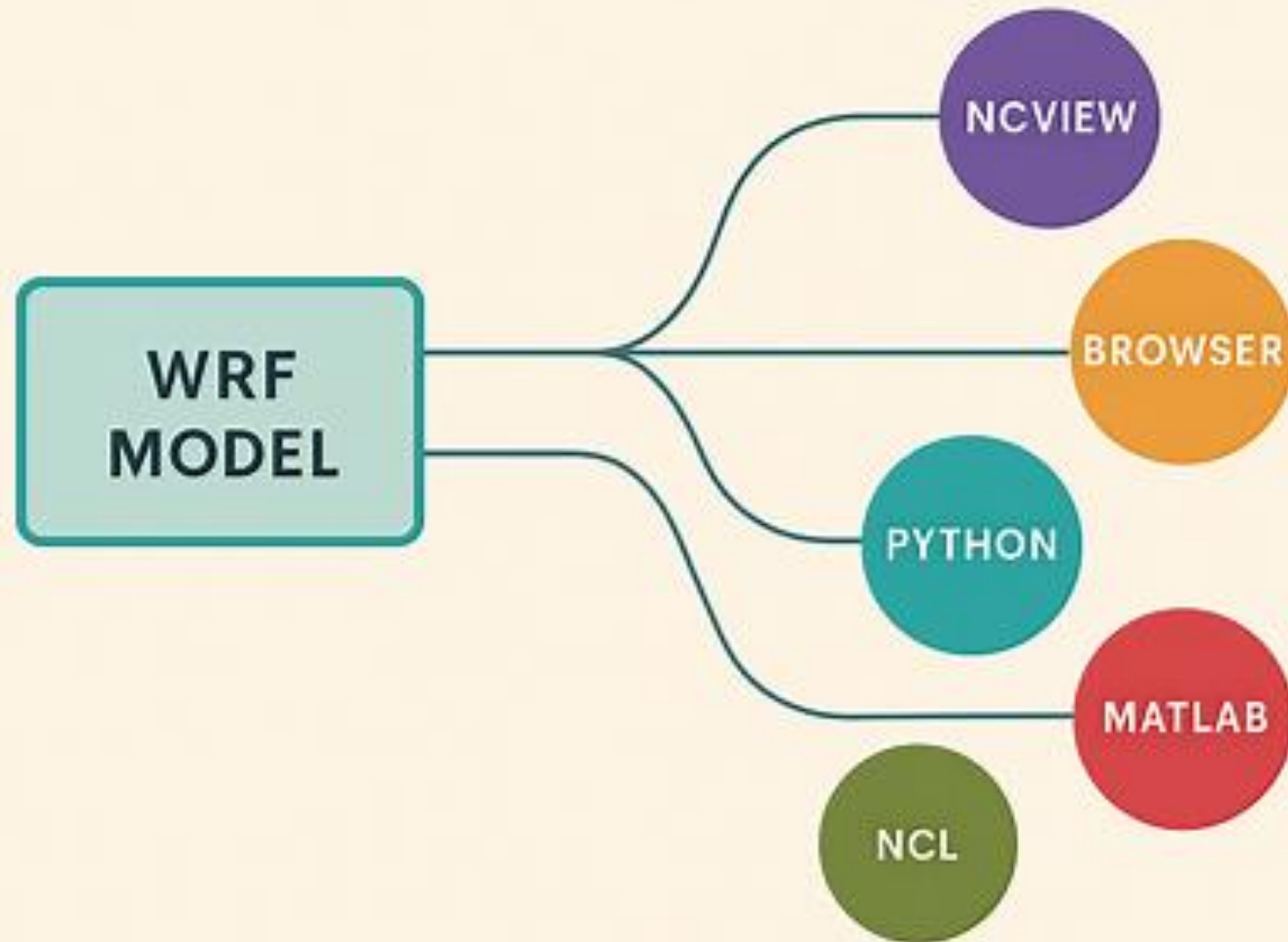
Model Execution



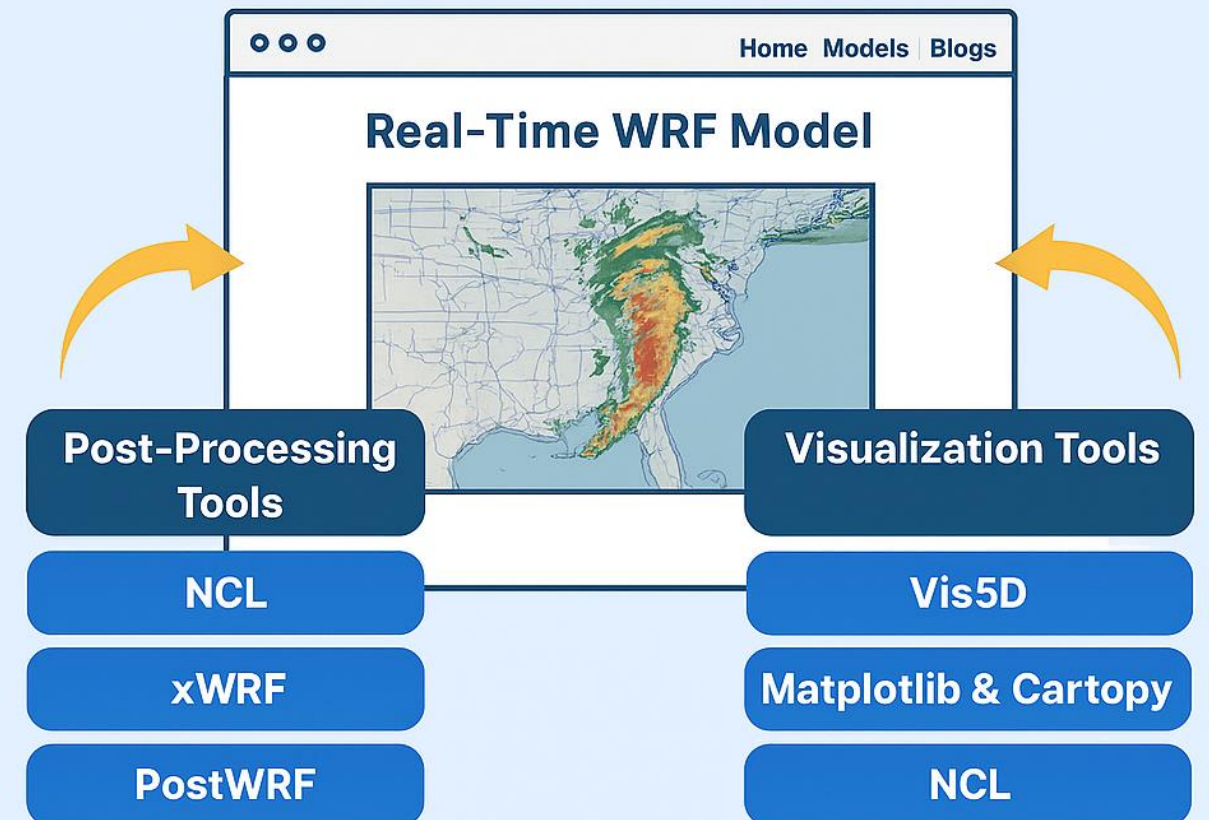
**Postprocessing &
Visualization**

Tools Used in Real-Time WRF Modelling System

Post-Processing and Visualisation of Real-Time WRF Modelling System



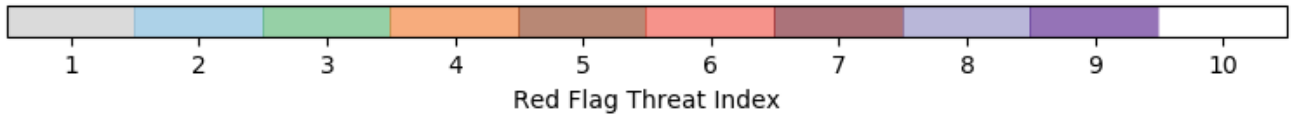
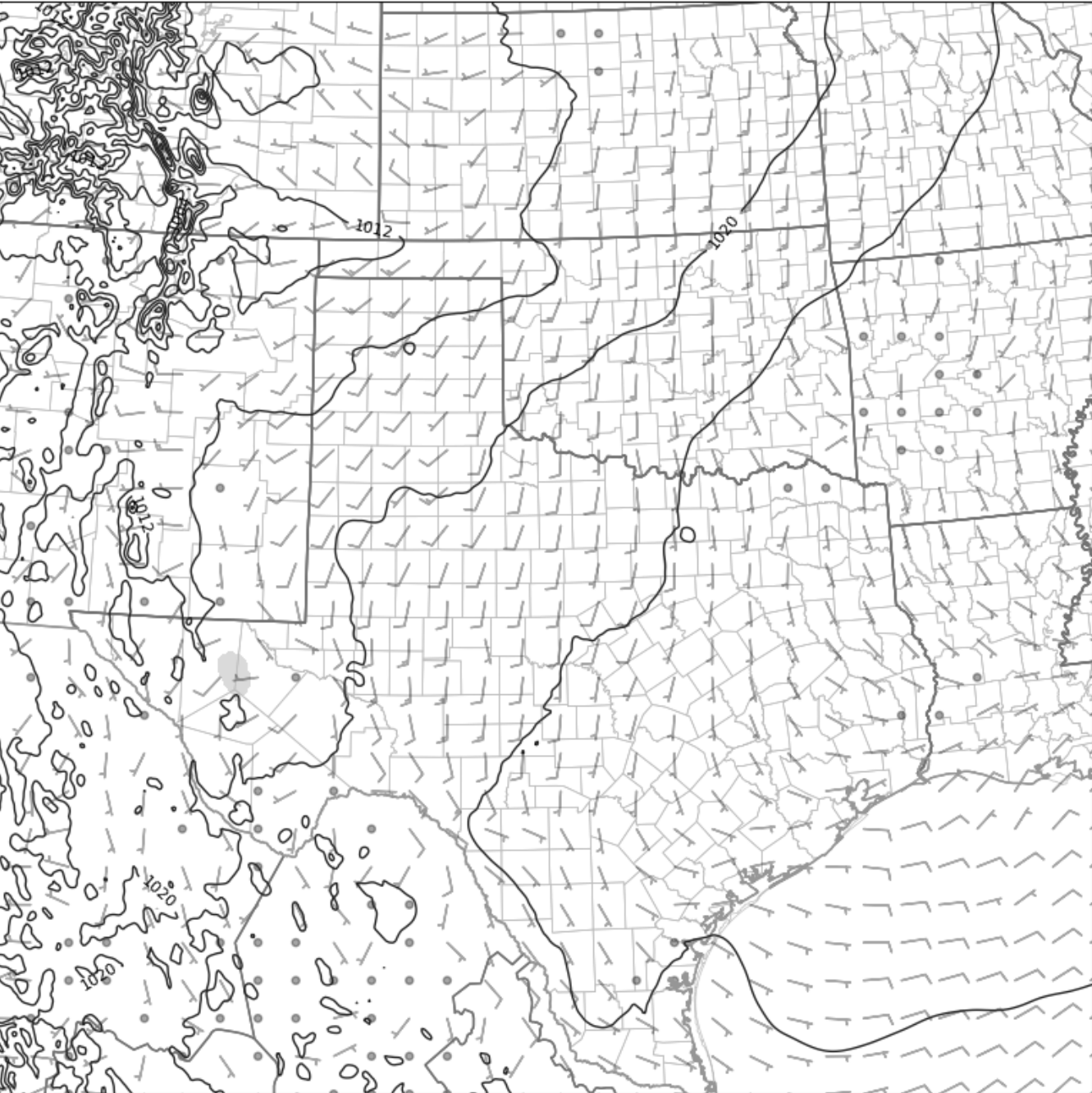
MODELING SYSTEM TOOLCHAIN



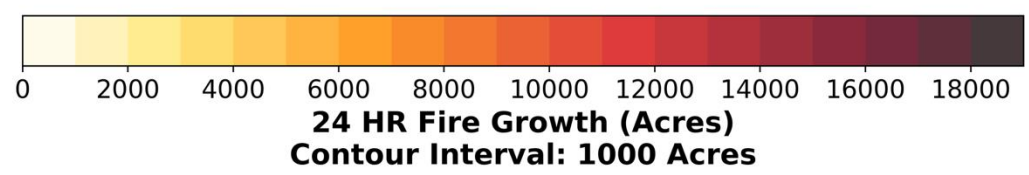
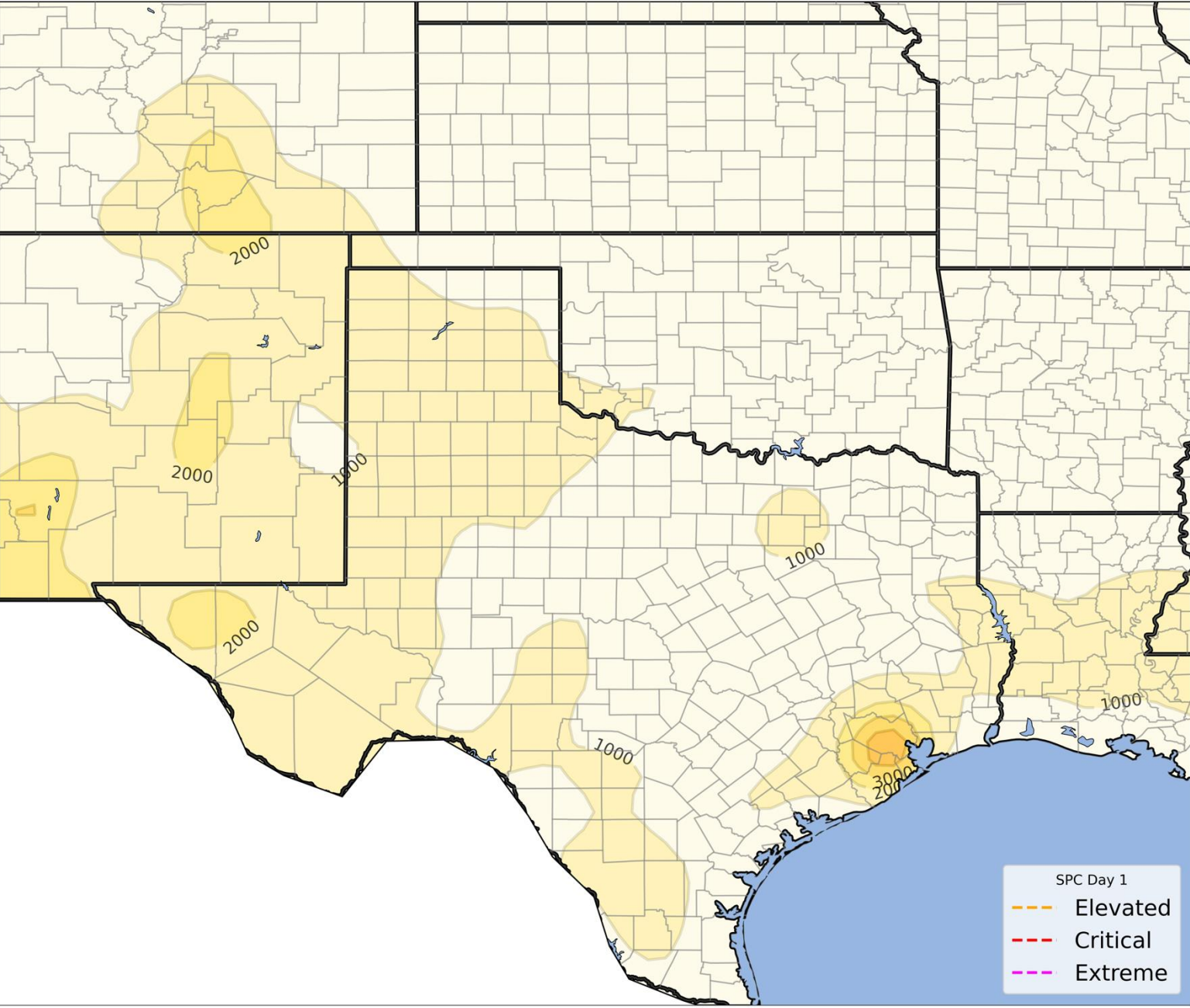
Realtime Fire Weather Forecast and Fire Risk Visualization

Red Flag Threat Index
Sea Level Pressure (mb), 10m Wind (barbs)

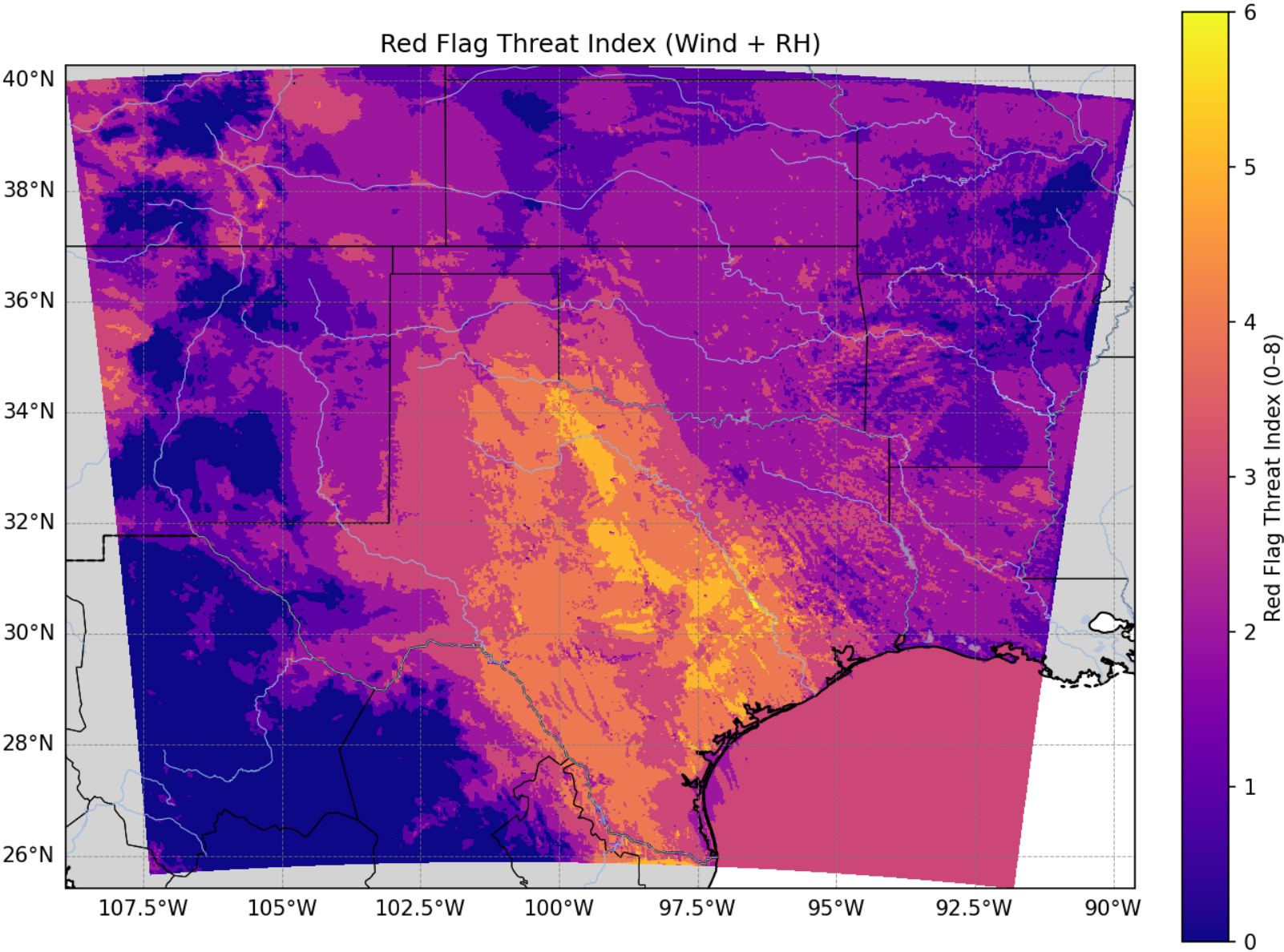
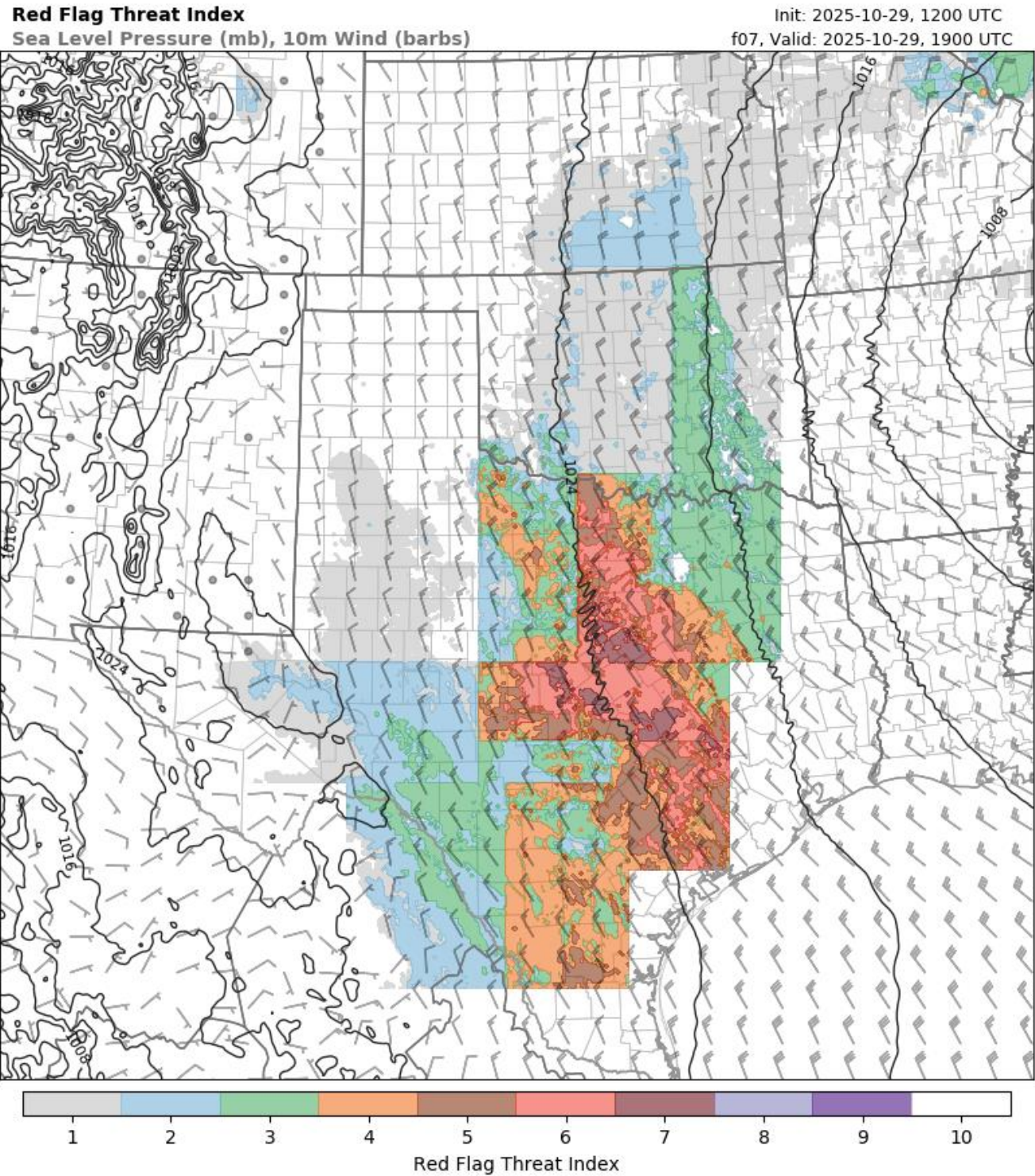
Init: 2025-11-04, 1200 UTC
f00, Valid: 2025-11-04, 1200 UTC



Forecasted 24-hr Fire Growth -- New Fires -- Valid: 11/04 18z - 11/05 18z (Day 1)



3km Real-Time RFTI System

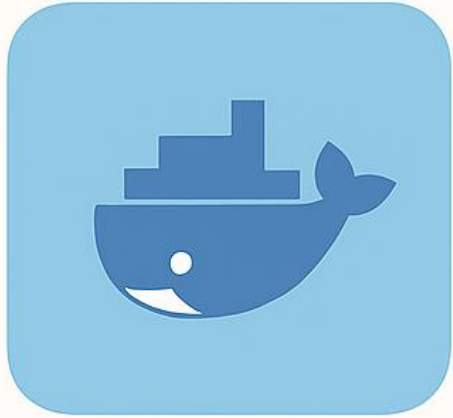


Data Integration with Other Sources

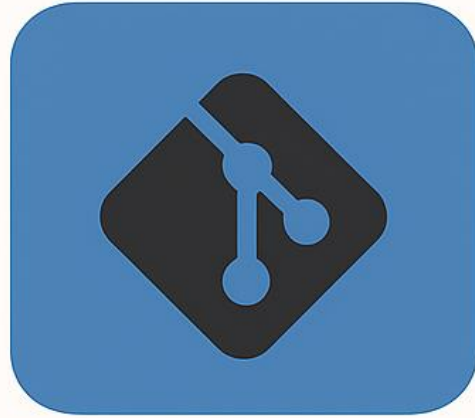
Processed data can be combined to maximize impact:

- MODIS / VIIRS (FIRMS) fire detection
- HRRR Smoke models (from NOAA)
- Land cover maps
- Population and Infrastructure data
- Ground air quality stations (e.g., EPA AirNow)

Approaches For Improving Reproducibility



Containerization



Version Control



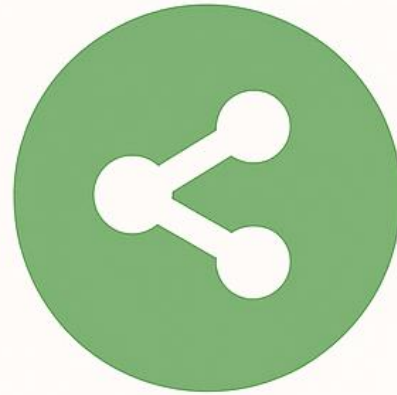
FAIR Principles



Workflow Automation



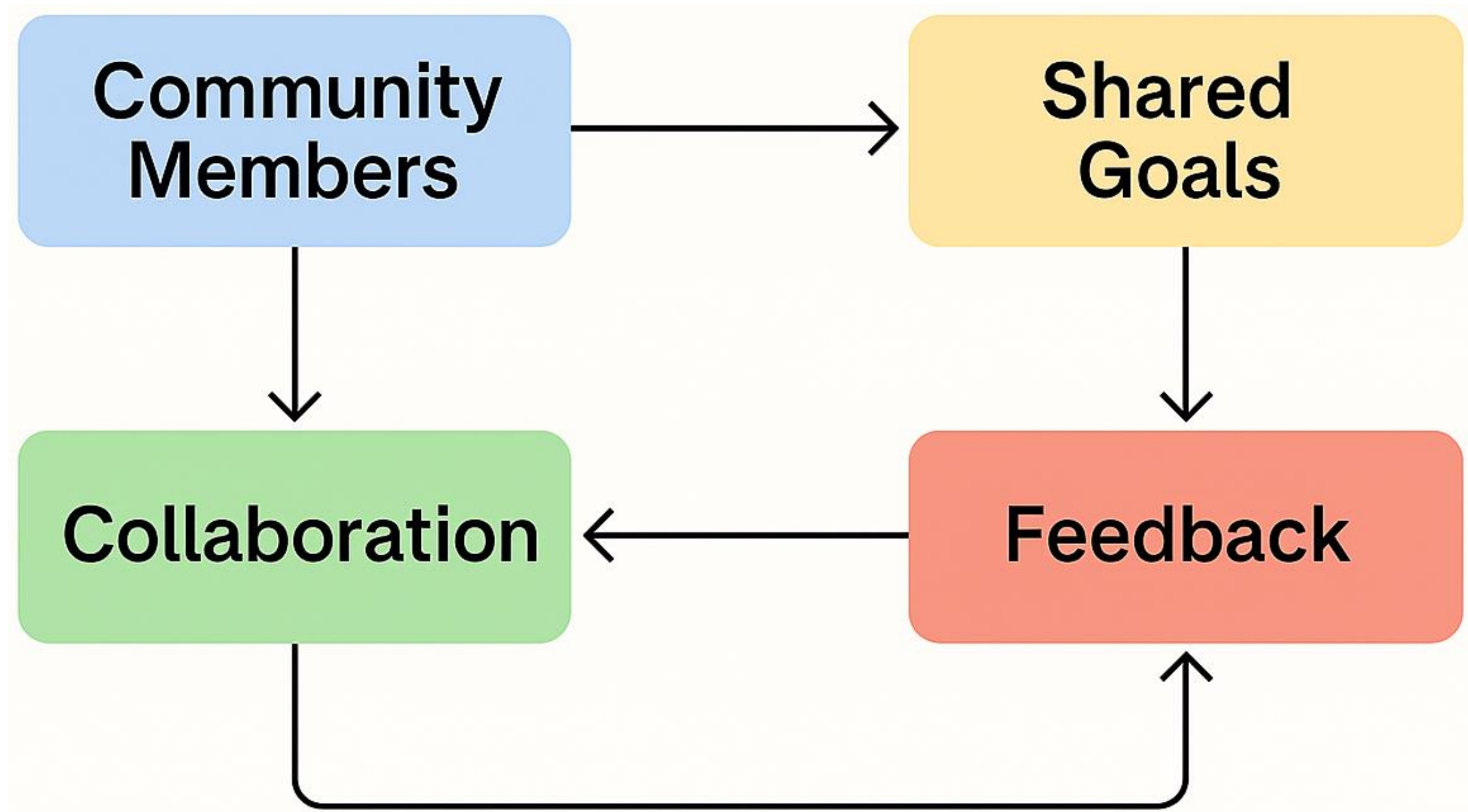
Metadata Standards



Open Data & Code Sharing

- **Containerization**
- Docker or Singularity to create portable environments.
- **Version Control**
- Git-based workflows for code and configuration tracking.
- **FAIR Principles**
- Ensure data is Findable, Accessible, Interoperable, and Reusable.
- **Workflow Automation** Tools like Snakemake or Nextflow
- **Metadata Standards**
- Comprehensive documentation for datasets and models.
- **Open Data & Code Sharing**
- Zenodo, GitHub, or ESGF for transparency.
- **Benchmarking & Validation**
- Standardized test cases for Earth system models.

Toward Community-Driven Reproducibility for Sustainable Science



Impact of Reproducibility on Decision Making



**Policy
Making**



**Disaster
Management**



**Long-Term
Climate
Planning**



Reproducibility

Conclusions and Future Pathways Toward Sustainable Science

Conclusions

- **Reproducibility is both a technical and cultural challenge** – requires robust tools and collaborative practices.
- **Promising tools and methodologies exist** – version control, FAIR principles, workflow automation.
- **Broader adoption and community-driven standards are essential** – shared benchmarks and transparent workflows.
- **Long-term impact** – strengthens scientific integrity and supports decision-making in climate policy and disaster management.

Future Pathways

- Extend reproducibility assessments to new Earth system frameworks.
- Integrate reproducibility metrics into software evaluation.
- Build reproducibility benchmarks for climate analytics pipelines.



Thank you!

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